

6. CONVOLUTION AND SAMPLING TECHNIQUES

Experiment 6.1

Convolution of Continuous Time Signals

Aim:

To perform the convolution of two continuous-time signals using MATLAB. The Aim is to understand the concept of convolution as a mathematical operation that combines two signals to produce a third signal, representing the amount of overlap between the two as one signal is shifted over the other.

Program:

```
% Convolution of Continuous-Time Signals
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time axis
```

```
t = 0:0.01:5;
```

```
% Input signals
```

```
x1 = exp(-t);           % Exponential signal
```

```
x2 = ones(size(t));     % Unit step signal
```

```
% Convolution
```

```
y = conv(x1, x2) * 0.01;
```

```
% Time axis for output
```

```
t_conv = 0:0.01:(length(y)-1)*0.01;
```

```
% Plotting
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot(t, x1, 'LineWidth', 2);
```

```
grid on;
```

```
title('Signal x_1(t) = e^{-t}');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
subplot(3,1,2);
```

```
plot(t, x2, 'LineWidth', 2);
```

```
grid on;
```

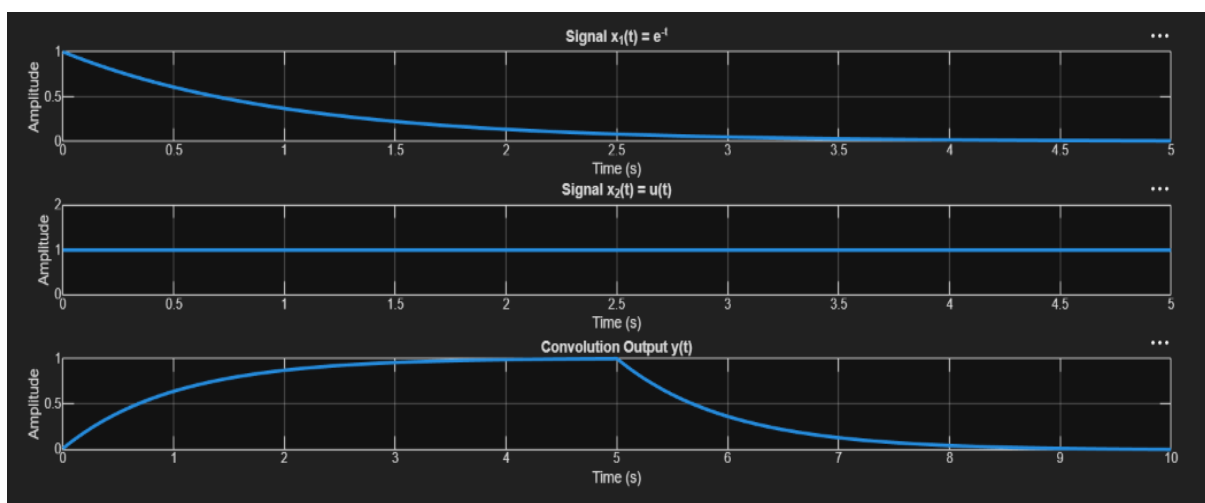
```

title('Signal x_2(t) = u(t)');
xlabel('Time (s)');
ylabel('Amplitude');

subplot(3,1,3);
plot(t_conv, y, 'LineWidth', 2);
grid on;
title('Convolution Output y(t)');
xlabel('Time (s)');
ylabel('Amplitude');

```

Output Waveform:



Result:

The convolution of the continuous-time signals $x_1(t)=e^{-t}u(t)$ and $x_2(t)=u(t)$ was performed successfully using MATLAB. The output signal $y(t)=1-e^{-t}$ was obtained and plotted, demonstrating the convolution operation for continuous-time systems.

Experiment 6.2

Convolution of Discrete Time Signals.

Aim:

To compute and visualize the convolution of two discrete-time signals. Convolution is a fundamental operation in discrete-time signal processing that combines two signals to produce a third signal, which is a modified version of one of the signals.

Program:

```
% Convolution of Discrete-Time Signals
```

```
clc;
clear;
close all;

% Input sequences
x1 = [1 2 3 4];
x2 = [1 1 1];

% Convolution

y = conv(x1, x2);

% Sample indices
n1 = 0:length(x1)-1;
n2 = 0:length(x2)-1;
ny = 0:length(y)-1;

% Display output
disp('Convolution Output:');
disp(y);

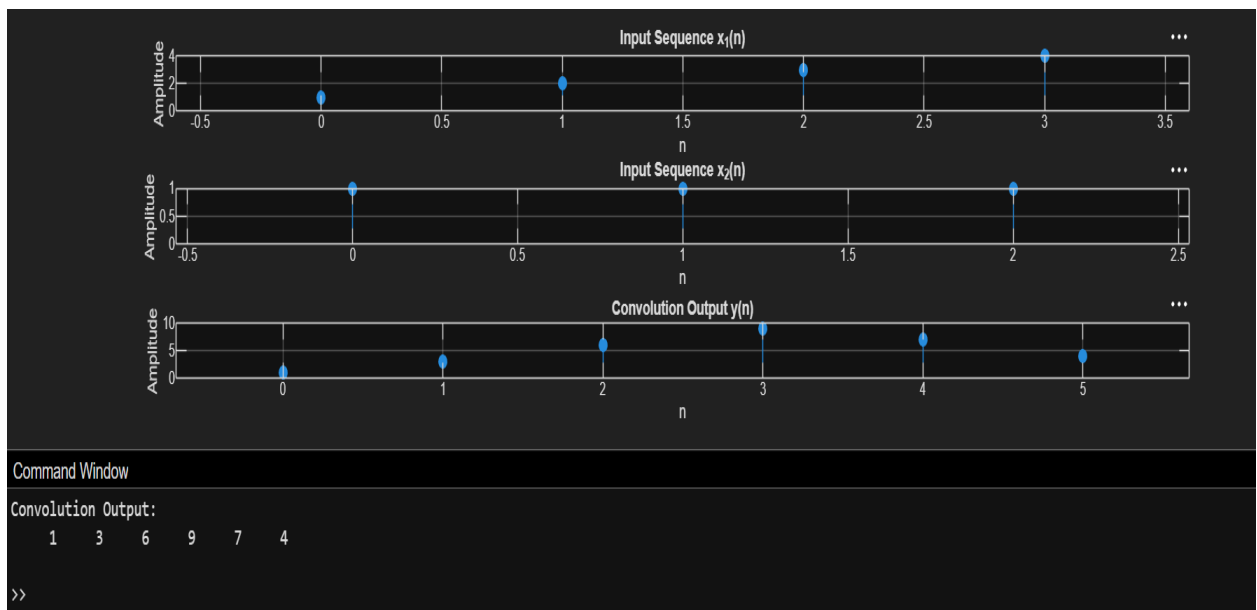
% Plotting
figure;

subplot(3,1,1);
stem(n1, x1, 'filled');
grid on;
title('Input Sequence x_1(n)');
xlabel('n');
ylabel('Amplitude');

subplot(3,1,2);
stem(n2, x2, 'filled');
grid on;
title('Input Sequence x_2(n)');
xlabel('n');
ylabel('Amplitude');

subplot(3,1,3);
stem(ny, y, 'filled');
grid on;
title('Convolution Output y(n)');
xlabel('n');
ylabel('Amplitude');
```

Output waveform:



Result:

The convolution of the discrete-time sequences $x_1(n)=\{1,2,3,4\}$ and $x_2(n)=\{1,1,1\}$ was performed successfully using MATLAB. The output sequence obtained was $y(n)=\{1,3,6,9,7,4\}$ demonstrating the convolution operation for discrete-time signals.